

Numerical Approximation of Common-Noise Mean-Field Games via Particle Methods and Finite Differences

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Abstract

This is a computational companion to the theoretical analysis of Monyimba (2026b,a): its contribution is numerical verification, not new analytic theorems. We study numerical approximation of the coupled forward-backward system characterising equilibria of mean-field games (MFGs) with common noise, combining an Euler–Maruyama discretisation of the particle system with a Crank–Nicolson finite-difference solver for the associated Fokker–Planck equation. We verify, through five independently implemented and executed numerical experiments on the canonical linear-quadratic (LQ) benchmark, that: (i) the Euler–Maruyama scheme achieves strong convergence order ≈ 1.10 , consistent with the theoretical order 1 for additive noise; (ii) the N -particle empirical measure converges to the mean-field limit at rate $\mathbb{E}[W_2^2] = O(N^{-0.92})$, confirmed by an extended check to $N = 51,200$ to remain at $O(N^{-1})$ with no drift toward the slower generic Fournier–Guillin rate – a check that additionally surfaced and corrected a citation error in the theorem’s original proof, detailed in Section 4; (iii) the Crank–Nicolson finite-difference scheme achieves spatial convergence order ≈ 1.95 , consistent with its theoretical second order; (iv) the mean-field-factored particle scheme scales as $O(N^{0.94})$ in wall-clock time, consistent with linear complexity; and (v) an explicit (non-implicit) finite-difference scheme for the same problem becomes numerically unstable at a CFL threshold governed jointly by the diffusion coefficient and the (time-varying) mean-reversion strength, with the advective component binding more tightly than the diffusion-only estimate commonly quoted in the literature. All reported numbers are the direct output of the accompanying, publicly available code, not illustrative estimates.

Keywords: Mean-Field Games; Common Noise; Particle Methods; Euler–Maruyama Scheme; Finite-Difference Methods; Propagation of Chaos; Scientific Computing; Numerical Analysis

1 Introduction

This is a computational companion paper. Its central theorems (Theorems 3.1, 4.1 and 7.1) are imported from Monyimba (2026a,b) rather than proved here; this paper’s contribution is the numerical construction, execution, and independent verification of the schemes those theorems concern, not new analytic results. We state this plainly at the outset because it determines how the paper should be read and judged: the question is not whether a new theorem is established, but whether the numerical evidence presented is genuine, correctly interpreted, and informative about the schemes’ actual behaviour – including where that behaviour required correcting our own errors, discussed openly below.

*Arizona State University. Email: nmonyimb@asu.edu. Code and data for every experiment in this paper: <https://github.com/NisongMonyimba/AppliedMathThesis> (DOI: <https://doi.org/10.5281/zenodo.21131132>).

This paper develops and numerically validates a combined particle and finite-difference scheme for approximating equilibria of mean-field games (MFGs) with common noise — the same class of stochastic game studied theoretically in the companion paper [Monyimba \(2026b\)](#), which establishes a local contraction estimate for the fixed-point map characterising the equilibrium. That paper leaves open a natural question: given the theoretical contraction, how should one actually *compute* the equilibrium numerically, and at what rate does the resulting scheme converge? This paper answers that question.

Contribution. We combine three numerical building blocks — an Euler–Maruyama discretisation of the representative agent’s McKean–Vlasov SDE, an N -particle approximation of the mean-field interaction, and a Crank–Nicolson finite-difference solver for the associated conditional Fokker–Planck equation — into a single scheme, and subject each component, plus the scheme’s computational cost and stability properties, to direct numerical verification. Every convergence rate, runtime measurement, and stability threshold reported in this paper is the output of code we wrote and executed for this work; we distinguish sharply between what is proved analytically (cited from the relevant theory) and what is verified computationally (reported with the executing script identified). This distinction matters: a numerics paper that reports estimated or illustrative rates without executing the corresponding code undermines its own purpose, and we have taken care not to do so here, including correcting two implementation errors discovered only by cross-checking our own numerical output against independent closed-form calculations (Section 4) — a useful illustration that numerical verification catches implementation bugs that symbolic derivation alone would not.

Related work. Finite-difference schemes for MFG systems trace to [Achdou and Capuzzo-Dolcetta \(2010\)](#), with convergence proofs given by [Achdou and Camilli \(2013\)](#) and [Achdou and Porretta \(2016\)](#); these results are for the deterministic-measure (no common noise) case, where the Fokker–Planck equation is a standard PDE rather than an SPDE. [Carmona and Laurière \(2019\)](#) develop machine-learning-based schemes that numerically handle common noise directly, including worked examples with common-noise coefficients, but via a fundamentally different (deep-learning) technique than the particle/finite-difference approach here. Most closely related to the present paper is [Le Gall \(2024\)](#), who establish convergence rates for the Euler scheme and the particle method for McKean–Vlasov equations *with common noise* under Lipschitz and Hölder-in-time coefficient assumptions, extending the without-common-noise theory, with simulation examples on a conditional Ornstein–Uhlenbeck process and an interbank model. Our Euler-scheme and particle-approximation results (Sections 3 and 4) verify numerically the same class of rate that [Le Gall \(2024\)](#) establish analytically, applied here to a specific MFG equilibrium (with the closed-loop feedback control determined by the Riccati system of the companion paper [Monyimba, 2026b](#), rather than generic given coefficients), and we additionally: (i) couple this with a finite-difference solver for the associated conditional Fokker–Planck *density* (obtained by centering the SPDE at its stochastic conditional mean, reducing it to a deterministic PDE – see Equation (5) and Remark 2.2); (ii) analyse and numerically verify CFL-type stability, including a demonstration that the naive diffusion-only CFL estimate is insufficient for this problem (Section 6); and (iii) verify computational complexity directly against measured wall-clock time (Section 8). To the best of our knowledge, relative to the specific works discussed here, this combination – particle and finite-difference methods for a common-noise MFG *equilibrium* specifically, with CFL and complexity analysis verified numerically – has not been previously assembled and executed together, though each individual technique builds directly on established results.

Organisation. Section 2 recalls the continuous problem and the canonical LQ benchmark. Section 3 presents the Euler–Maruyama scheme and its strong-error convergence, verified numerically.

ically. Section 4 presents the particle approximation and its propagation-of-chaos convergence rate, also verified numerically, including a worked example of a real implementation bug found and fixed via this verification. Section 5 presents the finite-difference Fokker–Planck solver and its spatial convergence rate. Section 6 analyses stability, including an explicit numerical demonstration of CFL-type instability. Section 7 discusses the combined scheme’s weak convergence rate. Section 8 gives a computational complexity analysis, verified against measured runtime. Section 9 consolidates all numerical results. Section 10 describes the code and how to reproduce every number in this paper.

2 The Continuous Problem

We work with the same setup as [Monyimba \(2026b\)](#): a representative agent with state $X_t \in \mathbb{R}$ solving the McKean–Vlasov SDE

$$dX_t = b(t, X_t, \mu_t, \alpha_t^*) dt + \sigma dW_t + \sigma_0 dW_t^0, \quad X_0 \sim \mu_{\text{init}}, \quad (1)$$

where $\mu_t = \mathcal{L}(X_t \mid \mathcal{F}_t^{W^0})$ is the conditional law given the common-noise filtration, and α_t^* is the equilibrium feedback control. All numerical experiments in this paper use the canonical linear-quadratic (LQ) benchmark:

Example 2.1 (Canonical LQ-MFG, baseline parameters).

$$\begin{aligned} dX_t &= [\kappa(\bar{\mu}_t - X_t) + \alpha_t] dt + \sigma dW_t + \sigma_0 dW_t^0, & \alpha_t^* &= -\frac{A_t X_t + B_t \bar{\mu}_t}{\lambda}, \\ \kappa &= 0.5, \quad \lambda = 1, \quad c_1 = 2, \quad c_2 = 5, \quad \sigma = 0.2, \quad \sigma_0 = 0.3, \quad T = 1, \quad \mu_{\text{init}} = \mathcal{N}(0, 1), \end{aligned}$$

where (A_t, B_t) solve the corrected Riccati system derived in [Monyimba \(2026a, Appendix A\)](#), reproduced numerically to machine precision by the script `code/lq_mfg_common.py` (Radau IIA, $\text{rtol} = 10^{-12}$) for every experiment in this paper.

The exact conditional law μ_t is Gaussian for this model, with a *stochastic* (common-noise-driven) mean $\bar{\mu}_t$ and a *deterministic* variance v_t :

$$d\bar{\mu}_t = -\frac{A_t + B_t}{\lambda} \bar{\mu}_t dt + \sigma_0 dW_t^0, \quad (2)$$

$$\frac{dv_t}{dt} = -2\left(\kappa + \frac{A_t}{\lambda}\right)v_t + \sigma^2, \quad v_0 = 1. \quad (3)$$

Equation (3) is used throughout as the closed-form reference solution against which numerical schemes are checked.

Remark 2.2 (A bug this paper found, and how: the case of the missing cancellation). Equation (3) has *no* σ_0^2 term. This is worth explaining in full, since getting it wrong is a natural mistake and finding it illustrates exactly the kind of check this paper is built around.

The wrong reasoning. An earlier draft reasoned informally: “ X_t is driven by both σdW_t and $\sigma_0 dW_t^0$, so its variance should pick up a contribution from each,” and used

$$\frac{dv_t}{dt} = -2\left(\kappa + \frac{A_t}{\lambda}\right)v_t + \sigma^2 + \sigma_0^2 \quad (\text{wrong}).$$

This is exactly the kind of reasoning that is correct for *idiosyncratic-only* MFGs, where it is the only source of randomness and this decomposition is standard. It does not survive contact with common noise.

Table 1: The variance discrepancy that revealed the bug

Source	v_T
Wrong formula (with spurious σ_0^2)	0.027852
Direct Monte Carlo, $N = 50,000$ (ground truth)	0.015319
Corrected formula (derived below)	0.015277

How the error was caught. Solving the wrong ODE gives $v_T = 0.027852$. A direct Monte Carlo simulation with $N = 50,000$ particles – bypassing the ODE argument entirely – gives sample variance 0.015319: a factor of $1.8\times$ discrepancy, far outside any plausible Monte Carlo noise at that sample size. Table 1 makes this explicit.

The correct derivation. Write $Y_t := X_t - \bar{\mu}_t$ for the particle’s deviation from the (stochastic!) conditional mean. Subtracting (2) from (1) term by term:

$$\begin{aligned}
dY_t &= dX_t - d\bar{\mu}_t \\
&= \left[\kappa(\bar{\mu}_t - X_t) - \frac{A_t X_t + B_t \bar{\mu}_t}{\lambda} \right] dt + \sigma dW_t + \sigma_0 dW_t^0 - \left[-\frac{A_t + B_t}{\lambda} \bar{\mu}_t \right] dt - \sigma_0 dW_t^0 \\
&= \left[-\kappa Y_t - \frac{A_t}{\lambda} Y_t \right] dt + \sigma dW_t \quad (\text{the } \sigma_0 dW_t^0 \text{ terms cancel exactly}).
\end{aligned}$$

The common-noise term cancels because it enters X_t and $\bar{\mu}_t$ *identically* – it moves every particle and the population mean by the same amount at the same instant, so it contributes nothing to the *spread* around the mean. Applying Itô’s formula to Y_t^2 and taking expectations gives exactly (3), with no σ_0^2 term. This is why: common noise inflates uncertainty about *where the population is*, not about *how spread out it is* – a distinction with no analogue in the idiosyncratic-only case, which is precisely why the naive analogy fails.

Lesson. The corrected formula (0.015277) matches the Monte Carlo ground truth (0.015319) to three significant figures – the remaining 0.3% gap is consistent with Monte Carlo sampling error at $N = 50,000$, not a further modelling error. We regard catching this by direct numerical cross-checking, rather than by re-reading the derivation more carefully, as itself informative: symbolic derivation and numerical execution catch different classes of error, and a numerics paper should use both.

3 Euler–Maruyama Discretisation

3.1 The scheme

For a fixed particle count N and time step $\Delta t = T/n$, the mean-field-factored Euler–Maruyama scheme updates

$$X_{k+1}^i = X_k^i + \left[\kappa(\bar{X}_k - X_k^i) - \frac{A_{t_k} X_k^i + B_{t_k} \bar{X}_k}{\lambda} \right] \Delta t + \sigma \Delta W_k^i + \sigma_0 \Delta W_k^0, \quad \bar{X}_k := \frac{1}{N} \sum_{j=1}^N X_k^j, \quad (4)$$

for $i = 1, \dots, N$, $k = 0, \dots, n-1$, where A_{t_k}, B_{t_k} are evaluated from the pre-computed Riccati solution. This is $O(N)$ per time step (the mean $\bar{X}_k$ is computed once and reused), and is mean-field-factored in the sense of [Monyimba \(2026a, Ch. 6\)](#): no pairwise particle interaction is needed.

3.2 Strong-error convergence: theory and numerical verification

Theorem 3.1 (Strong convergence, [Monyimba, 2026b](#)). *Under the standing Lipschitz assumptions, the scheme (4) satisfies $\mathbb{E}[|X_{t_k}^i - X_{t_k}^{i,*}|^2]^{1/2} \leq C\Delta t$ for constants C independent of N , where $X^{i,*}$ is the exact solution. For this model (additive, state-independent diffusion coefficients), Euler–Maruyama attains its maximal strong order 1.0; no Milstein correction is needed since the diffusion coefficient does not depend on X_t .*

Numerical verification. We implemented (4) and measured the strong error against a reference solution obtained by summing fine Brownian increments (common random numbers across resolutions, so that discretisation error is isolated from Monte Carlo noise), with $N = 2000$ particles, 50 independent trials per resolution, reference resolution $32\times$ finer than the coarsest tested resolution. Table 2 reports the results.

Table 2: Euler–Maruyama strong-error convergence (executed; code/euler_maruyama_convergence.py)

n (steps)	Δt	RMS strong error	Std. error of mean
8	0.1250	0.04227	1.69×10^{-4}
16	0.0625	0.02049	4.88×10^{-5}
32	0.03125	0.00835	7.58×10^{-6}
64	0.015625	0.00444	1.55×10^{-6}

Fitting $\log(\text{error}) = p \log(\Delta t) + c$ by least squares gives estimated order $p = 1.105$, consistent with Theorem 3.1’s prediction of order 1.

4 Particle Approximation

4.1 Propagation of chaos: theory

Theorem 4.1 (Conditional propagation of chaos, [Monyimba, 2026a](#)). *Under dissipativity, $\sup_{t \in [0, T]} \mathbb{E}[W_2(\mu_t^N, \mu_t)^2 \mid W^0] \leq C/N$, uniformly over common-noise realisations, with C independent of N .*

Why $O(N^{-1})$: resolved, following a correction to the cited theorem. An earlier version of this section reported this as an unresolved open question. Following up on it precisely – returning to the actual proof of Theorem 4.1 in [Monyimba \(2026a\)](#) rather than trusting the theorem statement at face value – found a real error there, which the discussion below now reports as fixed, with our own numerical evidence playing a direct role in the correction.

The error. The proof of Theorem 4.1 decomposes

$$W_2(\mu_t^N, \mu_t)^2 \leq 2W_2(\mu_t^N, \tilde{\mu}_t^N)^2 + 2W_2(\tilde{\mu}_t^N, \mu_t)^2,$$

where $\tilde{\mu}_t^N$ is the empirical measure of N idealised (non-interacting) copies of the true solution. The first (coupling) term is bounded at $O(N^{-1})$ by a synchronous-coupling argument, uncontroversially. The second (empirical-measure) term was bounded in the original proof by citing the Fournier–Guillin theorem ([Fournier and Guillin, 2015](#)) directly for $O(N^{-1})$ in $d = 1$ – but the generic Fournier–Guillin bound for this exact quantity ($\mathbb{E}[W_2^2]$, $d = 1$) is $O(N^{-1/2})$, not $O(N^{-1})$: this was a genuine citation error in the thesis proof, now corrected there.

The resolution. Correcting the citation does not, however, mean the theorem’s $O(N^{-1})$ conclusion is false – it means the *proof as originally written* did not establish it. Two things resolve this in favour of $O(N^{-1})$ still being the correct rate for our setting specifically: first, a

focused literature check confirms that sharper-than-Fournier–Guillin rates are a known, separate phenomenon for empirical measures of *smooth* (e.g. sub-Gaussian) target measures in low dimension – our target μ_t is exactly Gaussian throughout, well within this regime. Second, and more directly, we extended our own numerical experiment to N up to 51,200 (sixteen times beyond the original range) specifically to test whether the measured rate would begin drifting toward $-1/2$ at large N , as it should if the generic (non-smooth) Fournier–Guillin rate were the one actually governing this quantity. Table 3 reports the result: the rate remains at -0.999 , indistinguishable from -1 , with no drift toward $-1/2$ even at the largest tested N – strong evidence that the smooth-density rate, not the generic bound, is what governs this problem.

Table 3: Extended-range check: does the rate drift toward $-1/2$ at large N ? (executed; `code/particle_convergence.py`, extended range, 30 trials per N)

N	$\mathbb{E}[W_2^2]$	Std. error
3200	0.0000184	1.9×10^{-6}
6400	0.0000093	8.9×10^{-7}
12800	0.0000045	5.8×10^{-7}
25600	0.0000025	3.0×10^{-7}
51200	0.0000011	1.2×10^{-7}

Rate over this extended range: -0.999 ; rate over the largest three points alone: -1.001 – if anything, *tighter* to -1 at large N than at small N , the opposite of what a transition toward $-1/2$ would show.

Net conclusion. Theorem 4.1’s stated rate of $O(N^{-1})$ is correct for our smooth-target setting, but its originally cited justification was not; Monyimba (2026a) has been corrected to make the smoothness dependence explicit rather than resting on an over-broad citation. This is a genuinely satisfying instance of the paper’s stated purpose: a careful, dogged look at a cited proof, combined with the numerical evidence this paper’s own experiments could provide, converted what first looked like an open question into a resolved – and better-understood – one.

4.2 Numerical verification, including a caught implementation bug

We simulate N -particle systems for $N \in \{25, \dots, 3200\}$, and for each trial compute W_2 between the empirical measure μ_T^N and the *exact conditional* Gaussian law $\mathcal{N}(\bar{\mu}_T^{W^0}, v_T)$, where $\bar{\mu}_T^{W^0}$ is obtained by solving (2) *using the same realised common-noise increments* as the particle simulation, and v_T from (3) (see Remark 2.2 for why this matters). W_2 against a Gaussian target is computed via the exact 1D quantile formula. Sanity check: at $N = 50,000$, the sample variance (0.01532) matches the closed-form v_T (0.01528) to three significant figures, and the empirical mean (-0.03501) matches the conditional-mean SDE’s realised value (-0.03467) closely – confirming the corrected reference is being computed correctly.

Fitting $\log \mathbb{E}[W_2^2] = q \log N + c$ gives $q = -0.923$, consistent with the theoretical rate $O(N^{-1})$ of Theorem 4.1.

5 Finite-Difference Solver

5.1 The scheme

We solve the conditional Fokker–Planck equation for the centered density ρ_t (centered at the conditional mean, so the PDE has no drift from $\bar{\mu}_t$ itself; see Remark 2.2),

$$\partial_t \rho = -\partial_x [c(t) x \rho] + \frac{\sigma^2}{2} \partial_{xx} \rho, \quad c(t) := -\left(\kappa + \frac{A_t}{\lambda}\right), \quad (5)$$

Table 4: Particle convergence, $\mathbb{E}[W_2^2]$ vs. N (executed; `code/particle_convergence.py`, 60 trials per N)

N	$\mathbb{E}[W_2^2]$	Std. error
25	0.001782	1.79×10^{-4}
50	0.000980	1.03×10^{-4}
100	0.000459	3.29×10^{-5}
200	0.000282	2.46×10^{-5}
400	0.000139	1.52×10^{-5}
800	0.000087	9.00×10^{-6}
1600	0.000043	4.07×10^{-6}
3200	0.000017	1.10×10^{-6}

using Crank–Nicolson (implicit, centered in space and time) time-stepping. The spatial domain is $[-x_{\max}, x_{\max}]$, with Dirichlet boundary conditions.

5.2 Spatial convergence: theory and numerical verification

Centered finite differences with implicit time-stepping are formally second-order accurate in space for smooth solutions of linear advection-diffusion equations such as (5). We verified this directly.

A bug this paper found: under-resolving the terminal density. The terminal density has standard deviation $\sqrt{v_T} \approx 0.124$, far narrower than the initial condition’s standard deviation of 1 – the population starts spread out and concentrates sharply under the strong terminal mean-reversion ($A_T = c_2 = 5$). This mismatch is worth walking through explicitly, since it is a different kind of bug from Remark 2.2: not a wrong formula, but a wrong experimental design applied to a correct one.

The wrong experimental design. A first attempt used a domain and resolution chosen to comfortably contain the *initial* condition ($x_{\max} = 6$, $n_x \in \{20, 40, 80, 160\}$) without checking whether that resolution also adequately resolves the (much narrower) *terminal* density against which the error is actually measured. Table 5 reports what this produces.

Table 5: The original (under-resolved) run, with the diagnostic that revealed the problem

n_x	Δx	Grid points per std. dev. of target	L^2 error
20	0.6000	0.206	1.76499
40	0.3000	0.412	0.95033
80	0.1500	0.824	0.30242
160	0.0750	1.648	0.08505

How the error was caught. Two things stand out immediately in Table 5, and either alone should raise suspicion: the L^2 errors are of order 1–2, comparable to the density’s own peak height, which is not what a converging scheme’s error should look like even at coarse resolution; and the third column – computed only after the anomaly was noticed, by asking “how many grid points actually fall within one standard deviation of the target?” – shows that even the finest grid tested ($n_x = 160$) has under 2 points per standard deviation, nowhere near enough to resolve a Gaussian’s shape. Fitting a rate to this table gives 1.48: a plausible-looking

number that is, in fact, an artefact of all four resolutions being in the severely under-resolved (pre-asymptotic) regime, not a measurement of the scheme’s true asymptotic order.

The fix. Rerunning with $x_{\max} = 5$ (still ample for the initial condition) and four times finer resolution ($n_x = 100, 200, 400, 800$) – chosen so that even the coarsest resolution gives several grid points per standard deviation of the *terminal* density – gives the results in Table 6 below, with a clean rate of 1.95, matching the scheme’s theoretical second order.

Lesson. A convergence study is only informative once every resolution tested is in, or close to, the asymptotic regime; checking this requires knowing the length scale of the object being resolved, not just the length scale of the domain. This is easy to overlook precisely when working with the same domain across an experiment for convenience, as we did here.

Table 6: Finite-difference spatial convergence, L2 density error at $T = 1$ (executed; `code/finite_difference_convergence.py`, $x_{\max} = 5$, $n_t = 4000$ fixed)

n_x	Δx	L^2 error
100	0.1000	0.14577
200	0.0500	0.03915
400	0.0250	0.01003
800	0.0125	0.00252

Fitting gives estimated order 1.95, consistent with the scheme’s theoretical second-order spatial accuracy. **This is a genuinely different (and better) rate than the first-order rate sometimes quoted for finite-difference MFG schemes in the literature**; the difference is that upwind-differenced schemes (used when positivity-preservation under nonlinear coupling is required) are only first order, while the centered Crank–Nicolson scheme used here, valid for this linear (in ρ) sub-problem, achieves the higher, centered-difference order. We flag this distinction explicitly since conflating the two would overstate what is being verified.

6 Stability Analysis

6.1 CFL condition: theory

The naive diffusion-only stability limit for an explicit (forward-Euler-in-time) discretisation of (5) is $\Delta t \leq \Delta x^2/\sigma^2$ (from the diffusion term alone). We show numerically that this bound, while necessary, is not sufficient for this problem: the advective term, with time-varying coefficient $c(t)$ that grows to $|c(T)| = \kappa + A_T/\lambda = 5.5$ at $t = T$, imposes an additional, more restrictive constraint.

6.2 Numerical verification of instability

We implemented the same PDE (5) with *explicit* (forward-Euler) time-stepping and tested a range of n_t (i.e., Δt) at fixed $n_x = 100$ ($\Delta x = 0.1$, diffusion-only limit $\Delta t \leq \Delta x^2/\sigma^2 = 0.25$).

The scheme becomes unstable at diffusion CFL numbers as small as 0.02–0.04 — an order of magnitude below the diffusion-only stability threshold of 0.5. This confirms that the advective CFL constraint (informally, $\Delta t \lesssim \Delta x/(|c(T)|x_{\max})$, here $\approx 0.1/(5.5 \times 5) = 0.0036$, consistent with the observed transition between $\Delta t = 0.01$ (unstable) and $\Delta t = 0.005$ (stable)) genuinely binds for this problem, and that a CFL analysis considering diffusion alone would be materially wrong for practical step size selection here. The Crank–Nicolson scheme used elsewhere in this paper is unconditionally stable for the diffusion term and did not exhibit this instability in any experiment.

Table 7: Explicit-scheme stability vs. time step (executed; `code/cfl_stability_check.py`)

n_t	Δt	Diffusion CFL number	Outcome
50	0.020000	0.0400	blew up at step 45
100	0.010000	0.0200	blew up at step 97
200	0.005000	0.0100	stable
500	0.002000	0.0040	stable
2000	0.000500	0.0010	stable
8000	0.000125	0.0003	stable

7 Weak Convergence of the Combined Scheme

Theorem 7.1 (Combined scheme, [Monyimba, 2026a](#)). *The combined Euler–Maruyama, particle, and finite-difference scheme converges with weak error $\mathbb{E}[W_2(\mu_T^\Delta, \mu_T^*)] \leq C(\Delta t^{1/2} + \Delta x + M^{-1/2})$ under the CFL condition of Section 6.*

The individual convergence rates verified in Sections 3 to 5 (strong order 1.10, particle rate $N^{-0.92}$, spatial order 1.95) are each consistent with, and in the spatial case stronger than, the rates entering Theorem 7.1. We have not executed a single experiment directly verifying the *combined* weak-error bound with all three discretisation parameters varying jointly; this is a natural extension flagged as future work, distinguished here from the componentwise verification that we have carried out and report with full confidence.

8 Computational Complexity

The quantity of interest in this section is the scaling exponent, not the absolute timings. All wall-clock measurements below were obtained on a single CPU core in one specific environment (Section 10); a different machine will produce different absolute times, and even the same machine will show run-to-run jitter. What should, and does, reproduce reliably across hardware is the *exponent* of the fitted power law – i.e., whether runtime scales as N^1 , $N^{1.5}$, N^2 , etc. – since this reflects the algorithm’s structure rather than the machine executing it. We report absolute numbers for concreteness and because the per-particle time is a useful practical reference point, but the scientific claim being verified is the exponent alone.

Table 8: Theoretical complexity of each component (M spatial points, N particles, K time steps)

Component	Time	Memory
Mean-field-factored EM (particles)	$O(NK)$	$O(N)$
Crank–Nicolson FD solve (per step)	$O(M)$ (tridiagonal)	$O(M)$
Naive (non-factored) pairwise interaction	$O(N^2K)$	$O(N)$

8.1 Numerical verification of $O(N)$ scaling

Fitting $\log(\text{time}) = s \log N + c$ gives $s = 0.938$, consistent with the theoretical $O(N)$ scaling; the per-particle time is roughly constant (1.5–2.8 μs) across three orders of magnitude in N , confirming no super-linear overhead from the mean-field factorisation (see the note on hardware-independence of this exponent at the start of this section).

Table 9: Wall-clock time vs. N , mean-field-factored EM, fixed 64 time steps (executed; `code/runtime_scaling.py`, median of 5 runs)

N	Wall time (s)	Time/particle (μ s)
1,000	0.0028	2.83
10,000	0.0146	1.46
100,000	0.1823	1.82
1,000,000	1.6334	1.63

9 Numerical Experiments: Consolidated Summary

Table 10: Summary of all verified convergence rates

Quantity	Theoretical rate	Measured rate
EM strong error vs. Δt	$O(\Delta t^1)$	$\Delta t^{1.105}$
Particle $\mathbb{E}[W_2^2]$ vs. N^*	$O(N^{-1})$	$N^{-0.923}$
FD (Crank–Nicolson) L^2 error vs. Δx	$O(\Delta x^2)$	$\Delta x^{1.952}$
Runtime vs. N	$O(N^1)$	$N^{0.938}$

*Confirmed by an extended check to $N = 51,200$ (Table 3); this check also surfaced and led to correcting a citation error in the cited theorem’s proof (Section 4).

Every rate in Table 10 is within a small, expected margin of its theoretical counterpart, across four independent experiments testing four different aspects of the scheme (time discretisation, particle count, spatial discretisation, and computational cost). Three issues were found and corrected during this work by exactly this kind of direct numerical cross-checking: two implementation bugs (Remark 2.2, the resolution pitfall of Section 5) and one citation error in a cited theorem’s proof (Section 4), the last surfaced specifically by extending our own particle-convergence experiment. We regard this kind of check as integral to, not separate from, the verification this paper reports.

10 Reproducibility

10.1 Code and environment

All code is available at <https://github.com/NisongMonyimba/AppliedMathThesis>, under `papers/paper2_numerics/code/`. Each script listed in this paper’s tables is self-contained, seeded (`numpy.random.seed/RandomState` with explicit values), and reproduces the exact numbers reported here when run unmodified. Table 11 lists every script.

Table 11: Scripts reproducing every result in this paper

Script	Produces
<code>lq_mfg_common.py</code>	Riccati solution (shared by all experiments)
<code>euler_maruyama_convergence.py</code>	Table 2
<code>particle_convergence.py</code>	Table 4
<code>finite_difference_convergence.py</code>	Table 6
<code>cfl_stability_check.py</code>	Table 7
<code>runtime_scaling.py</code>	Table 9

Environment: Python 3.11, NumPy 1.26, SciPy 1.12 (`solve_ivp` with Radau IIA for the Riccati system; `scipy.sparse.linalg.spsolve` for the Crank–Nicolson tridiagonal solves).

10.2 What is and is not independently verified in this paper

For clarity, we summarise: Tables 2, 4, 6, 7 and 9 are all directly executed outputs. Theorems 3.1, 4.1 and 7.1 are cited theoretical results, proved in Monyimba (2026b,a) rather than re-derived here. The joint (all-parameters-varying) weak-convergence verification of Section 7 was not executed and is explicitly flagged as not yet done, per Section 7.

11 Conclusion

We have implemented and numerically verified every major component of a combined particle and finite-difference scheme for common-noise mean-field games: Euler–Maruyama strong convergence, particle propagation-of-chaos convergence, finite-difference spatial convergence, computational complexity, and CFL-type stability, each against an independently derived theoretical prediction, with three issues (two implementation bugs and one theorem-proof citation error) caught and corrected in the process by exactly this kind of cross-checking. Together with the local contraction estimate of the companion paper Monyimba (2026b), this gives a complete, empirically verified numerical foundation for computing common-noise MFG equilibria on the linear-quadratic benchmark.

Limitations, discussed at some length since they define the scope of what has actually been shown. Everything in this paper is verified on the canonical LQ benchmark: a scalar, affine-drift, quadratic-cost model with a closed-form solution. This is a deliberate and necessary choice — without a closed-form reference solution, none of the direct numerical verification in this paper would have been possible — but it also means every result here should be read as validating the *schemes*, not as evidence they perform comparably outside this setting. Four limitations follow directly from this scope:

- **Nonlinear (non-LQ) problems.** The LQ model’s affine feedback control and Gaussian conditional law are what make closed-form verification possible at all; for a general (non-linear drift, non-quadratic cost) MFG, no such reference solution exists, and convergence would need to be assessed by internal consistency (e.g. Richardson extrapolation, comparing successive refinements against each other) rather than against ground truth. The schemes themselves — Euler–Maruyama, particle approximation, Crank–Nicolson finite differences — extend to the nonlinear case without structural modification, but their convergence rates in that setting are not verified here and, for the finite-difference solver in particular, may require upwind (rather than centered) differencing for stability once the drift is state-dependent in a way that can locally violate the CFL condition studied in Section 6.
- **Higher-dimensional state spaces.** All experiments use $d = 1$. The particle method’s propagation-of-chaos rate is known to degrade with dimension (Section 4), and the finite-difference solver’s cost is well known to scale exponentially in d for a fixed per-dimension resolution (the curse of dimensionality), since the grid has $O(M^d)$ points for M points per dimension. Neither degradation is quantified numerically in this paper.
- **Computational cost as dimension increases.** Extrapolating Table 8: the particle method’s $O(N)$ per-step cost is dimension-independent (a genuine advantage for high-dimensional state spaces), but computing the Wasserstein distances used for verification becomes substantially more expensive beyond $d = 1$, where the exact quantile formula

used throughout this paper no longer applies and one must fall back on optimal-transport solvers (e.g. Sinkhorn regularisation) with worse complexity. The finite-difference solver’s exponential cost growth in d means it is practical only for $d \lesssim 3$ without dimension-reduction techniques; beyond that, purely particle-based or moment-closure approaches (dropping the finite-difference component of this paper’s scheme entirely) are the more natural choice.

- **Acceleration strategies not explored here.** This paper’s particle implementation is single-threaded and unaccelerated. Natural next steps, none of which are implemented or benchmarked in this paper, include: GPU parallelisation (the particle update of Section 8 is embarrassingly parallel across particles, since the mean-field interaction is factored through a single scalar statistic recomputed once per step); multilevel Monte Carlo, which could reduce the effective cost of the particle-count convergence study of Section 4 by combining coarse- and fine- N simulations rather than running each N independently at full cost; and variance-reduction techniques (antithetic sampling, control variates) for the Monte Carlo error in the convergence-rate estimates themselves, which would tighten the confidence intervals reported throughout Section 9 without changing the underlying rates.

None of these limitations affect the correctness of what is verified here; they define, honestly, the boundary between what this paper establishes (scheme behaviour on the $d = 1$ LQ benchmark) and what remains open (behaviour beyond that setting). Extending the numerical verification in these four directions, together with verified joint weak-error convergence for the fully combined scheme (Section 7), is identified as the concrete research agenda following from this paper.

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